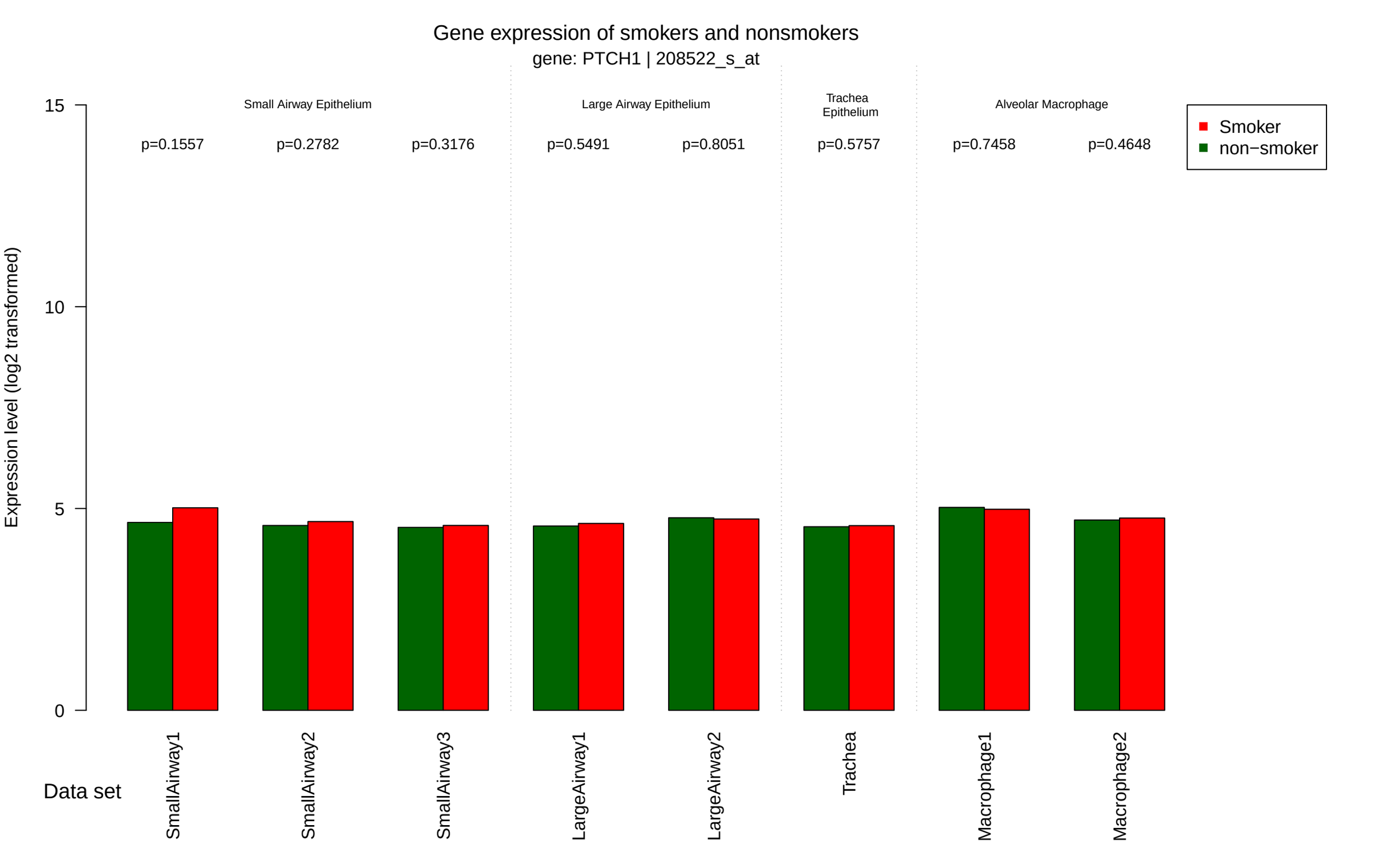
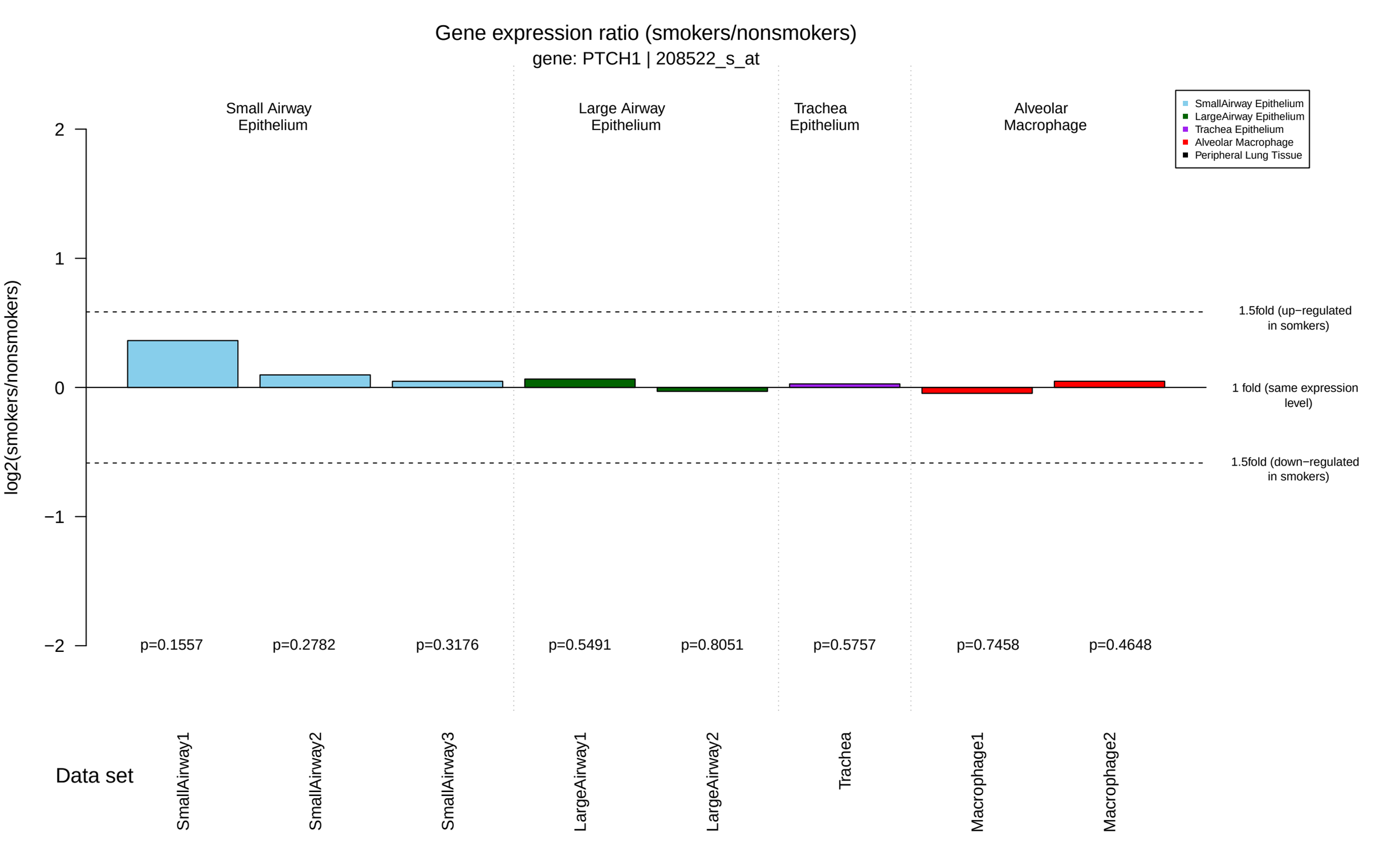
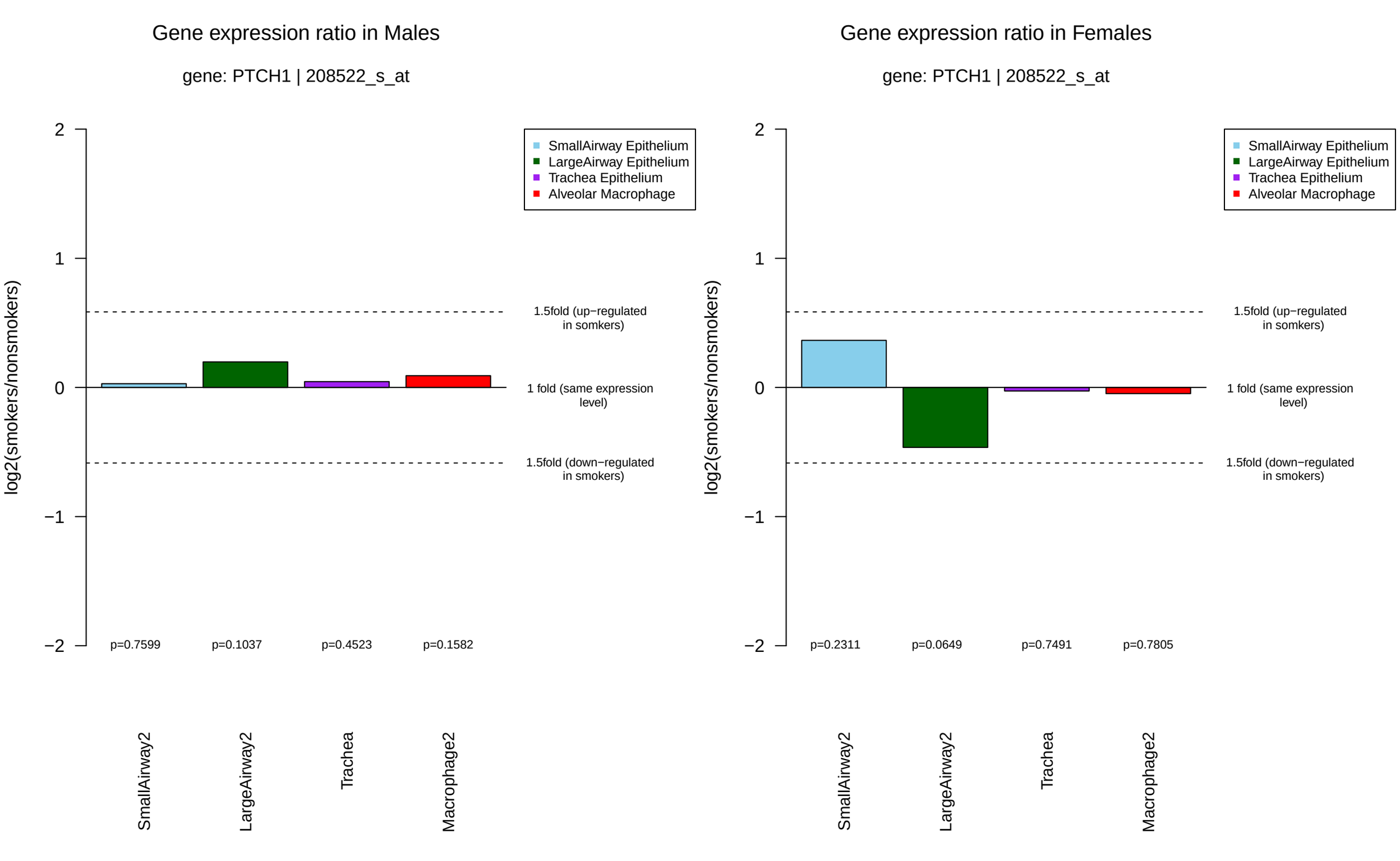
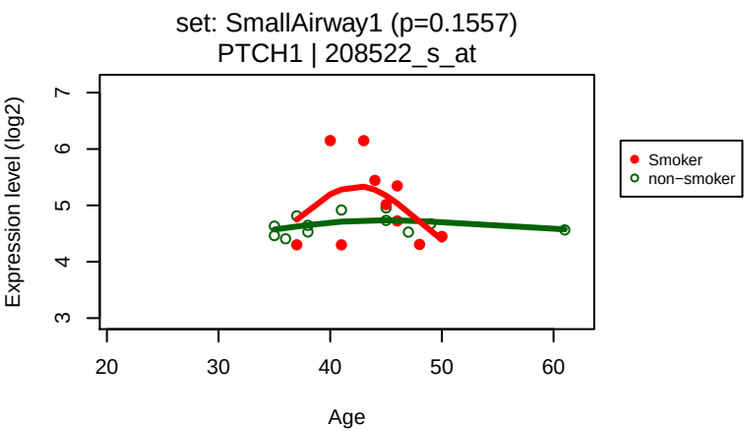
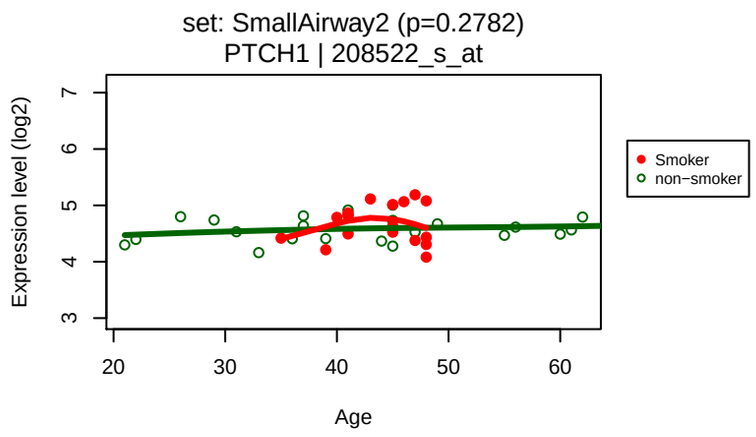
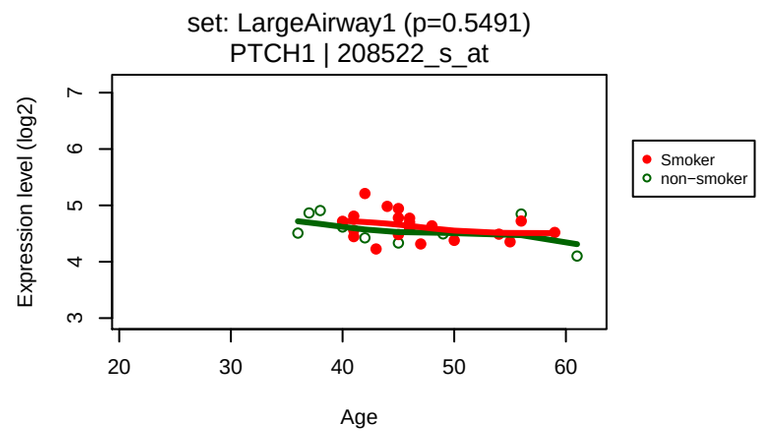

Gene expression of smokers and nonsmokers
gene: PTCH1 | 208522_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1557
p=0.2782
p=0.3176
p=0.5491
p=0.8051
p=0.5757
p=0.7458
p=0.4648
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

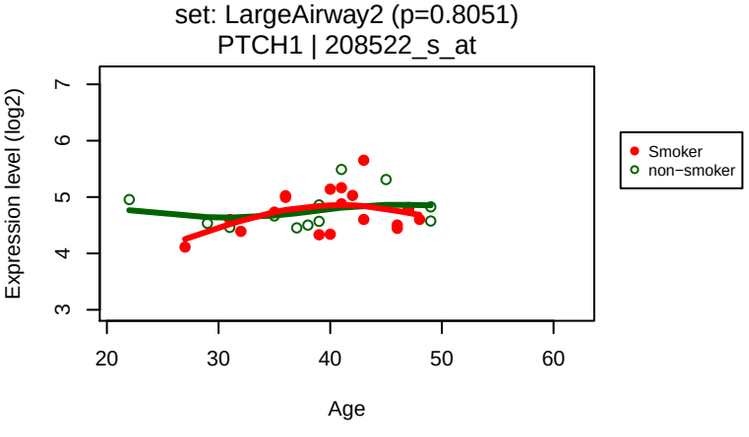

Gene expression ratio (smokers/nonsmokers)
gene: PTCH1 | 208522_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1557
p=0.2782
p=0.3176
p=0.5491
p=0.8051
p=0.5757
p=0.7458
p=0.4648
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

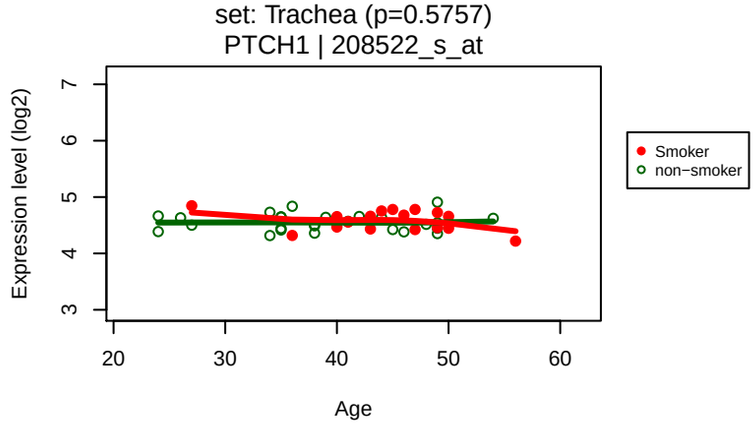

Gene expression ratio in Males
gene: PTCH1 | 208522_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7599
p=0.1037
p=0.4523
p=0.1582
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTCH1 | 208522_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2311
p=0.0649
p=0.7491
p=0.7805
SmallAirway2
LargeAirway2
Trachea
Macrophage2

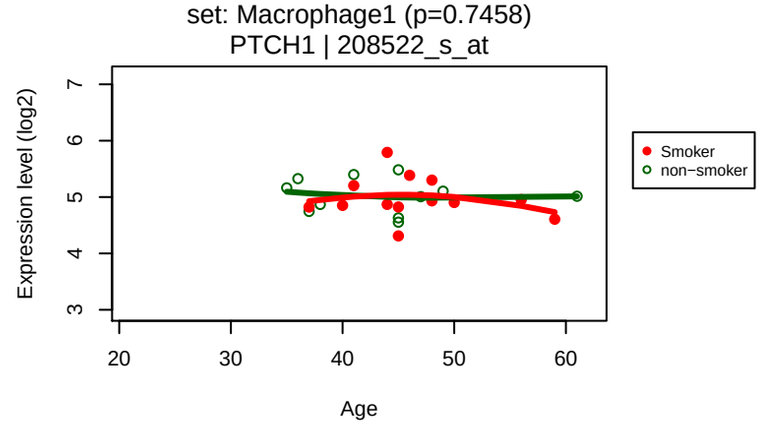

set: SmallAirway1 (p=0.1557)
PTCH1 | 208522_s_at
Expression level (log2)
Age
Smoker
non−smoker

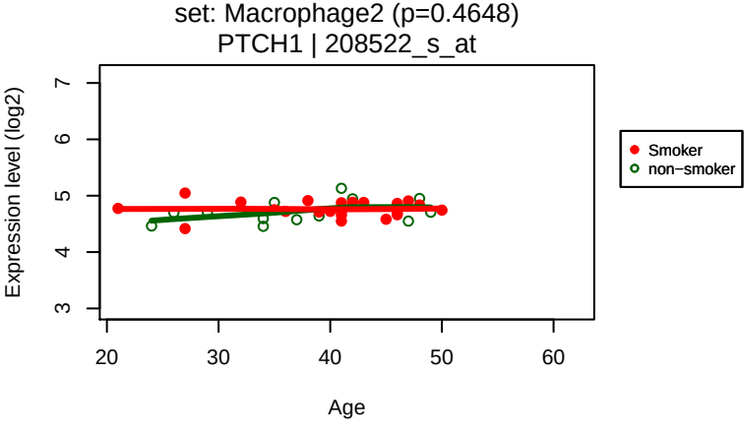

set: SmallAirway2 (p=0.2782)
PTCH1 | 208522_s_at
Expression level (log2)
Age
Smoker
non−smoker

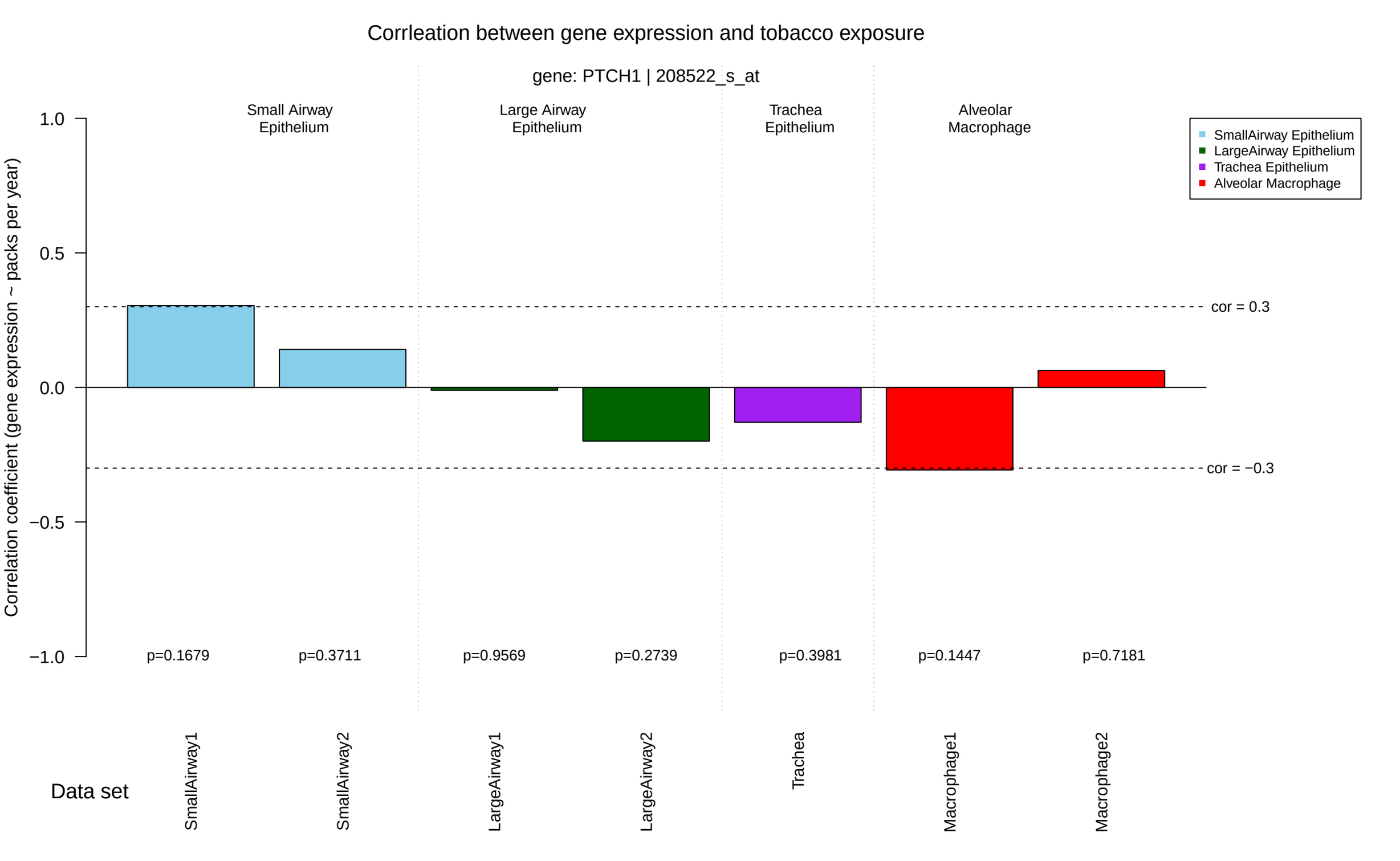

set: LargeAirway1 (p=0.5491)
PTCH1 | 208522_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8051)
PTCH1 | 208522_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5757)
PTCH1 | 208522_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7458)
PTCH1 | 208522_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4648)
PTCH1 | 208522_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTCH1 | 208522_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1679
p=0.3711
p=0.9569
p=0.2739
p=0.3981
p=0.1447
p=0.7181
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2